SoftWhere: Differentiable Selection Done Right

A sampling-free, multi-foveal TokenLearner selector for LookWhere — proposal with preliminary de-risking results

1. Background and Motivation

- **LookWhere** trains its selector by distilling the DINOv2 teacher's last-layer attention into a 2D importance map, then takes a hard top-k. Selection is non-differentiable, so the extractor's loss does not flow back to *where* to look.
- The authors tried to make the selection differentiable end-to-end by testing Gumbel-top-k (with a straight-through estimator) and REINFORCE (Table 12, Fig. 7b). Both of these methods hurt performance, and plain attention distillation won. They give a precise reason for this: training fails to converge without the map-distillation loss, which is a KL regularizer toward the teacher's "policy." All of their differentiable configurations already included that regularizer, but the extractor-signal gradient on top of it still did not help.
- **TokenLearner** calculates soft spatial attention in a fully differentiable way that avoids sampling altogether. Instead of relying on Gumbel or action sampling, it generates S distinct attention maps, meaning every token focuses on a different area.

2. The Gap and the Bar to Beat

The honest distinction is **sampling-based vs. sampling-free**, not “high-variance vs. low-variance.” LookWhere's Gumbel-top-k is already the low-variance reparameterized option used to avoid REINFORCE's variance, and it still failed (it was the worst at k=16/72). So variance alone is not the explanation. What both their methods share is **stochastic sampling of the selection** (plus, for Gumbel-ST, straight-through bias on the selection distribution). A TokenLearner selector removes the sampling entirely while keeping the mandatory KL regularizer. Because even the low-variance Gumbel-ST failed, a clean result here is itself diagnostic.

Baselines (LookWhere Table 12; ViT-S, kNN top-1; cls = ImageNet-HR class token, seg = ADE20K patch tokens). SoftWhere must clear the bolded row:

Selector-map training	cls k=16	cls k=72	cls k=128	seg k=16	seg k=72	seg k=128
none (stop-grad distill) — winner	**50.9**	**63.0**	**66.5**	**51.1**	**60.2**	**62.9**

Gumbel-top-k (straight-through)	44.5	61.4	66.2	49.1	58.9	62.5
REINFORCE	47.9	61.8	66.0	50.7	59.6	62.8

Note the gaps **collapse at k=128** (66.5/66.2/66.0; 62.9/62.8) — the clearest room to win is at k=16 and k=72.

3. Hypothesis

A soft, distillation-regularized TokenLearner selector is as good as or better than LookWhere's hard top-k selector. It succeeds where Gumbel-top-k and REINFORCE failed because it is differentiable without needing stochastic sampling, which is a trait both of those older methods share. Additionally, its multi-foveal maps can help improve coverage on multi-object images.

4. Method

Selector. Replace LookWhere's per-patch MLP SelectorHead with a TokenLearner that emits S spatial attention maps over the low-res selector grid (11×11 at 154-px input). The S maps are aggregated (max / mean / log-sum-exp union) into one importance map and upsampled to the high-res selector-map resolution, reusing LookWhere's existing interpolation machinery.

Selection / fetch — two operating modes:

- **(a) Soft-to-hard:** aggregate the S maps into a differentiable importance map; hard-fetch top-k high-res patches at its peaks; gradients flow through the soft aggregate via a straight-through estimator on the *aggregate* (not per-patch Bernoullis). Sampling-free but not bias-free.
- **(b) Soft low-res pathway:** keep maps soft for the low-res representation and hard-fetch high-res patches only at peaks — the genuinely sampling- and straight-through-free path. Lead the “deterministic” claim with mode (b).

Losses. L = λ_cls·L_cls + λ_pat·L_pat + λ_map·L_map(soft-aggregate-map, teacher attention). L_map is retained as the KL regularizer LookWhere found essential, now applied to the differentiable TokenLearner aggregate, with extractor gradients also flowing to the selector. An optional diversity term across the S maps discourages collapse.

5. Preliminary Results

We implemented SoftWhere end-to-end against the public LookWhere-DINOv2 checkpoint and ran a feasibility spike. Full ImageNet pretraining was out of scope; the goal was to de-risk the mechanism, produce artifacts, and stress-test the claims.

5.1 The mechanism is feasible and gradients flow

The **TokenLearner** selector head drops into **LookWhere** and produces a valid selector map (both v1.0/sigmoid and v1.1/softmax). We verified that gradients flow from an extractor-feature loss, through the hard top-k via a straight-through gate (gate_st = gate_hard + (gate_soft − gate_soft.detach())), and back to the **TokenLearner** parameters. All of the selector-head parameters receive non-zero, finite gradients, and there is zero leakage into the frozen backbone or extractor. This mechanically demonstrates the core claim of differentiable selection without sampling.

Known limitation: the straight-through top-k only re-weights currently-selected patches; it cannot directly pull in an excluded patch. A perturbed / soft top-k is future work.

5.2 Distillation recovers and decomposes the teacher saliency

By freezing the backbone and extractor, we can train just the TokenLearner head to match the pretrained LookWhere saliency (KL over spatial positions). This setup converges in minutes, dropping the KL from 0.15 to 0.056 on 16 images. The aggregate is a match for the teacher map, and the S maps break it down into distinct foveal regions. When tested on 200 held-out images, the distilled selector agrees with the teacher's top-k at a recall of 0.42 and an IoU of 0.27 (compared to 0.10 for random chance). This shows that the model successfully generalizes beyond the initial 16 distillation images.

Figure 1. *Distilled SoftWhere (v1.0). Left to right: input; LookWhere's single map; SoftWhere aggregate (recovers the teacher); the four distinct foveal maps.*

5.3 The multi-foveal maps are genuinely distinct (no collapse)

Sweeping the diversity regularizer (v1.0, S=4, 100 images): the maps are already distinct unregularized, and the knob drives pairwise overlap to 0.02 at essentially no fidelity cost. div=1 is the sweet spot.

diversity weight	fidelity KL ↓	pairwise overlap ↓	diversity ↑	teacher agreement ↑
0	0.094	0.188	0.812	0.456
1 (sweet spot)	**0.094**	**0.021**	**0.979**	**0.434**
2	0.102	0.070	0.930	0.418

5.4 v1.0 (sigmoid) vs v1.1 (softmax)

Data for the design fork: v1.0 fits the teacher better and agrees more, so we lead with it; v1.1 is "naturally multi-foveal" (softmax-over-space forces the maps to compete, so they are diverse even unregularized) but at a fidelity cost.

variant	fidelity KL ↓	diversity (unregularized)	teacher agreement ↑
v1.0 (sigmoid)	0.094	0.81	0.434
v1.1 (softmax)	0.144	0.96	0.392

5.5 Multi-object coverage on ADE20K — a cautionary negative result

We built an object-level coverage harness on ADE20K val (2000 images + masks): object-level recall via connected components, restricted to genuine small objects (3–25 patches) on multi-object images, at k_ratio 0.10. Using the distilled head (distilled in-domain on ADE20K train):

selector	object-coverage recall ↑
LookWhere single map	**0.701**
random	0.606
SoftWhere aggregate	0.468
SoftWhere multi-foveal (top-k/S per map)	0.364

Finding: The distilled multi-foveal head does not actually improve coverage. In fact, it covers fewer distinct small objects than LookWhere, and even fewer than a random selection. There are a couple of reasons for this. First, the TokenLearner map is natively 11×11 and bilinearly upsampled to 37×37. This is much coarser than the MLP head's super-resolved 44×44 map, meaning it cannot properly localize small objects. Second, its top-k selection is spatially clustered into blurry blobs, hitting very few distinct objects compared to how a random selection scatters. The main takeaway is that map diversity does not necessarily mean the object coverage will spread out.

Interpretation. This keeps the claim honest. The theoretical coverage advantage depends on end-to-end, coverage-driven training, which is precisely the goal of this proposal. It will also likely require a higher-resolution selector map. Distillation alone does not provide that for free. We have built and validated the harness and metric (scoring 0.61 for random versus 0.70 for LookWhere), so we are ready for the actual experiment.

Note: Check out this repository to reproduce the results from above (https://github.com/engichang1467/softwhere)

6. Proposed Experiments

Mirror LookWhere's exact selector-ablation protocol for direct comparability:

- **Classification:** kNN on ImageNet-HR, class token, leave-one-out (Table 12 / Fig. 6–7).

- **Segmentation:** kNN on ADE20K patch tokens (Table 12).
- **k settings:** k ∈ {16, 72, 128}, exactly as the paper.
- **Coverage:** object-level recall on ADE20K (the harness from §5.5), single-map vs. multi-foveal, after end-to-end training.
- **Ablations:** S (number of foveal maps); aggregation (union/max/mean); regularizer weight λ_map; diversity weight; v1.0 vs. v1.1; mode (a) vs. (b); selector-map resolution.

7. Risks and Mitigations

Risk	Mitigation
Reproduces the negative result (soft selection also fails)	This is still a contribution. Because the low-variance Gumbel-ST already failed, using a deterministic estimator without sampling allows us to separate three hypotheses: sampling noise, straight-through bias, and inductive bias. LookWhere's two previous attempts actually confounded those factors.
Selector-map resolution (NEW, from §5.5)	The 11×11 TokenLearner map is coarser than the MLP head's super-resolved 44×44 and hurts small-object localization. Apply the existing super-resolution rearrange to each foveal map, or emit maps at higher resolution; ablate resolution explicitly.
Map diversity does not yield coverage spread (NEW, from §5.5)	Train end-to-end with the extractor signal (and optionally a selection-level coverage term), not distillation alone; the spike shows distillation-to-teacher reproduces single-mode concentration.
Soft aggregation collapses to a single mode	Not observed in the spike (§5.3): maps are distinct even unregularized; the diversity term drives overlap to 0.02 at no fidelity cost.
Beating a published negative result is hard	Frame as a rigorous re-examination; the coverage analysis and the sampling-free diagnosis stand on their own even if accuracy ties.

8. Success Criteria

SoftWhere performs as well as or better than stop-grad distillation on kNN classification and segmentation at k values of 16, 72, and 128. Specifically, it achieves classification scores of at least 50.9, 63.0, and 66.5, and segmentation scores of at least 51.1, 60.2, and 62.9, along with a measurable gain in multi-object coverage. A clean negative result that diagnoses sampling noise, straight-through bias, or inductive bias is also an acceptable secondary outcome.

9. Compute Budget

For the selector ablations, LookWhere used ViT-S and trained for 100 to 400 epochs on ImageNet-1k. This setup is tractable on a single A100 for each configuration, although multiple GPUs speed up the full sweep. There is no teacher update. As in LookWhere pretraining, the teacher's forward pass accounts for most of the per-step cost. We ran the feasibility spike (§5) on a single MIG A100 slice, and each run took only a few minutes.

References

A. Fuller et al. *LookWhere? Efficient Visual Recognition by Learning Where to Look and What to See from Self-Supervision.* NeurIPS 2025. arXiv:2505.18051 (Table 12 and Fig. 6–7, selector-training ablations).

M. Ryoo et al. *TokenLearner: What Can 8 Learned Tokens Do for Images and Videos?* NeurIPS 2021. arXiv:2106.11297.